$\boxed{\text{HW} - 8}$

① $(1-x^2)\,y'' + 2xy' + 2y = 0$

$P(x) = \dfrac{2x}{1-x^2}$; $Q(x) = \dfrac{2}{1-x^2}$.

$x = \pm 1 \longrightarrow$ Regular singular points $\longrightarrow$ Poles of order 1.

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y'(x) = \sum_{n=0}^{\infty} (n+r)\, a_n x^{n+r-1}$$

$$y''(x) = \sum_{n=0}^{\infty} (n+r)(n+r-1)\, a_n x^{n+r-2}$$

Substituting into the ODE:

$$\sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} - \sum_{n=0}^{\infty} (n+r)(n+r-1)\, a_n x^{n+r}$$

$$+ 2\sum_{n=0}^{\infty} (n+r)\, a_n x^{n+r} + 2\sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

The lowest power of x : x^{n-2} @ $n=0$. Coefficient of x^{n-2} must vanish for indicial eqn:

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$\therefore \quad r(r-1) = 0 \Rightarrow \boxed{r = 0, 1}$

$\underline{r_1 = 0}$:

$$\underbrace{\sum_{n=2}^{\infty} n(n-1)\, a_n x^{n-2}}_{[n-2 \to n]} - \sum_{n=2}^{\infty} n(n-1)\, a_n x^n + 2 \sum_{n=1}^{\infty} n a_n x^{n-0} + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

[Term is 0 for $n=0$]

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)\, a_{n+2} x^n - \underbrace{\sum_{n=0}^{\infty} n(n-1)\, a_n x^n}_{\text{[Terms are 0 for } n=0,1]} + 2 \sum_{n=0}^{\infty} n a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = 0.$$

$\Rightarrow$

$\Rightarrow$ $\sum_{n=2}$ comparing coefficient of x^n of both sides:

$$(n+2)(n+1)\, a_{n+2} + a_n [-n(n-1) + 2n + 2] = 0.$$

$$\Rightarrow \quad a_{n+2} = \frac{a_n [n^2 - n - 2n - 2]}{(n+1)(n+2)}$$

$$\boxed{a_{n+2} = \frac{a_n (n^2 - 3n - 2)}{(n+1)(n+2)}}$$

$\underline{n=0}$: $\quad a_2 = \frac{-2a_0}{2} = -a_0$

$\underline{n=1}$: $\quad a_3 = \frac{a_1(-4)}{2 \cdot 3} = -\frac{2a_1}{3}$

$\underline{n=2}$: $\quad a_4 = \frac{a_2(-4)}{3 \cdot 4} = -\frac{a_2}{3} = \frac{a_0}{3}$

$\vdots$

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 \left(1 - x^2 + \frac{x^4}{3} + \dots\right) + a_1 \left(x - \frac{2x^3}{3} + \dots\right)$$

$\underline{\text{Even soln}}$: $\quad a_0 = 1 \; ; \; a_1 = 0.$

$$y_1(x) = a_0 \left(1 - x^2 + \frac{x^4}{3} + \dots\right)$$

Odd Sol^n: $a_0 = 0$; $a_1 = 1$.

$$y_2(x) = x - \frac{2x^3}{3} + \dots$$

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$y_1, y_2 \longrightarrow$ 2 LD sol^{n}s.

$$\therefore \boxed{y(x) = a_0 \quad c_1 y_1(x) + c_2 y_2(x)}$$ (Ans)

Region of validity : $|x| < 1$.

(2) (a) $y'' + \omega^2 y = 0$.

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} n a_n x^{n-1}$$

$$y''(x) = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$$

[Terms are 0 for $n=0$ & $n=0,1$].

$\therefore$ Substituting into the ODE :

$$\underbrace{\sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}}_{[n-2 \to n]} + \omega^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \left[(n+1)(n+2) a_{n+2} + \omega^2 a_n\right] x^n = 0$$

Comparing both sides:

$$(n+1)(n+2) a_{n+2} + \omega^2 a_n = 0.$$

$$\Rightarrow \boxed{a_{n+2} = \frac{-\omega^2 a_n}{(n+1)(n+2)}}$$

$n=0$: $a_2 = -\frac{\omega^2 a_0}{2} = -\omega^2 a_0/2!$

$n=1$: $a_3 = \frac{-\omega^2 a_1}{6} = -\omega^2 a_1/3!$

$n=2$: $a_4 = -\frac{\omega^2 a_2}{3 \cdot 4} = +\frac{\omega^4 a_0}{24} = \frac{\omega^4 a_0}{4!}$

$n=3$: $a_5 = -\frac{\omega^2 a_3}{4 \cdot 5} = +\frac{\omega^4 a_1}{4 \cdot 5 \cdot 6} = \frac{\omega^4 a_1}{5!}$

$$\therefore y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0\left(1 - \frac{\omega^2 x^2}{2!} + \frac{\omega^4 x^4}{4!} + \dots\right)$$

$$+ \frac{a_1}{\omega}\left(\omega x - \frac{(\omega x)^3}{3!} + \frac{(\omega x)^5}{5!} + - \dots\right)$$

$$\Rightarrow \boxed{y(x) = a_0 \cos\omega x + \frac{a_1}{\omega} \sin\omega x} \quad \text{(Ans)}$$

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(c) $y(x) = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} c_n x^{n+r}$

$$\Rightarrow y'(x) = \sum_{n=0}^{\infty} (n+r)\, c_n x^{n+r-1}$$

$$\Rightarrow y''(x) = \sum_{n=0}^{\infty} (n+r)(n+r-1)\, c_n x^{n+r-2}$$

$\therefore$ Substituting into the ODE:

$$\sum_{n=0}^{\infty} (n+r)(n+r-1)\, c_n x^{n+r-2} + \omega^2 \sum_{n=0}^{\infty} c_n x^{n+r} = 0.$$

Lowest power of x: x^{r-2} @ $n=2$. For indicial eq^n, coefficient of x^{r-2} must vanish!

$\therefore$ $r(r-1) = 0 \Rightarrow \boxed{r = 0, 1}$

$$\therefore \sum_{n=0}^{\infty} \left[(n+r+2)(n+r+1)\, c_{n+2} \cancel{x^{n+r}} + \omega^2 c_n\right] x^{n+r} = 0$$

$[n-2 \to n]$

Comparing both sides:

$$(n+r+2)(n+r+1)\, c_{n+2} = -\omega^2 c_n$$

$$\Rightarrow \boxed{c_{n+2} = \frac{-\omega^2 c_n}{(n+r+2)(n+r+1)}}$$

We solved the eq^n for $r=0$. Doing the same for $r=1$.

$$\boxed{c_{n+2} = \frac{-\omega^2 c_n}{(n+2)(n+3)}}$$

$n=0$: $\cancel{c_2 = \frac{-\omega^2 c_0}{2\cdot 3}}$

$n=1$: $\cancel{c_3 = \frac{-\omega^2 c_1}{3\cdot 4}}$

$n=2$: $\cancel{c_4 = \frac{-\omega^2 c_2}{4\cdot 5}}$

$\vdots$

$n=n$: $\cancel{c_{n+2} = \frac{-\omega^2 c_n}{(n+2)(n+3)}}$

~~Multiplying~~,

$n=0$: $c_2 = \frac{-\omega^2 c_0}{2\cdot 3} = \frac{-\omega^2 c_0}{3!}$

$n=1$: $c_3 = \frac{-\omega^2 c_1}{3\cdot 4} = \cancel{\frac{-\omega^2 c_1}{4!}}$ $\frac{-2\omega^2 c_1}{4!}$

$n=2$: $c_4 = \frac{-\omega^2 c_2}{4\cdot 5} = \frac{+\omega^4 c_0}{5!}$

$n=3$: $c_5 = \frac{-\omega^2 c_3}{5\cdot 6} = \frac{2\omega^4 c_1}{6!}$

$$\therefore \; y(x) = \sum_{n=0}^{\infty} c_n x^{n+1}$$

$$= x\left[c_0\left(1 - \frac{\omega^2 x^2}{3!} + \frac{\omega^4 y^4}{5!} - \dots\right) + c_1\left(x - \frac{2\omega^2 x^3}{4!} + \frac{2\omega^4 x^5}{6!} - \dots\right)\right]$$

$$= c_0\left(x - \frac{\omega^2 x^3}{3!} + \frac{\omega^4 x^5}{5!} - \dots\right) + c_1\left(x^2 - \frac{2\omega^2 x^4}{4!} + \frac{2\omega^4 x^6}{6!} + \dots\right)$$

If we write $y(x) = \sum_{n=0}^{\infty} c_n x^{n+1}$

$$y'(x) = \sum_{n=0}^{\infty} (n+1) c_n x^n$$

$$y''(x) = \sum_{n=0}^{\infty} (n+1) n \, c_n x^{n-1}$$

$$\therefore \sum_{n=0}^{\infty} n(n+1) c_n x^{n-1} + \sum_{n=0}^{\infty} \omega^2 c_n x^{n+1} = 0.$$

$\Rightarrow$ $\sum_{n=0}^{\infty}$ Coefficient of x^0: $2c_1 = 0 \Rightarrow \boxed{c_1 = 0}$

$$\therefore \; y(x) = \frac{c_0}{\omega}\left(\omega x - \frac{\omega^3 x^3}{3!} + \frac{(\omega x)^5}{5!} - \dots\right) = \frac{c_0}{\omega} \sin(\omega x)$$

This is the sine-part of the sol^n for part (a). The other part (cosine-part) can be obtained by carrying out the same analysis for $r=0$ & setting $\Leftrightarrow$ $\boxed{c_1 = 0}$.

(4) $x^2 y'' + 4xy' + (x^2+2)y = 0$

$$P(x) = \frac{4}{x} \; ; \quad Q(x) = \frac{x^2+2}{x^2}.$$

$x=0 \longrightarrow$ Regular singular pt.

$P(x) \longrightarrow$ Pole of order 1.
$Q(x) \longrightarrow$ Pole of order 2] @ $x=0$

(a): $y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$.

$$y'(x) = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} \; , \quad y''(x) = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

HW - 8 (PH 3103)

(Math Methods)

SOLUTIONS

(2)

(4) (a) Substituting into the ODE:

(1)

$$\sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r} + 4 \sum_{n=0}^{\infty} (n+r) a_n x^{n+r}$$

$$+ \underbrace{\sum_{n=0}^{\infty} a_n x^{n+r+2}}_{n \to n-2} + 2 \sum_{n=0}^{\infty} a_n x^{n+r} = 0.$$

$\Rightarrow$

$$\Rightarrow \sum_{n=2}^{\infty} \left([(n+r)(n+r-1) + 4(n+r) + 2] a_n + a_{n-2} \right) x^{n+r}$$

$$+ a_0 r(r-1) x^r + 4 r a_0 x^r + 2 a_0 x^r + \qquad \text{— (1).}$$

$$a_1 (1+r) r x^{r+1} + (1+r) a_1 x^{r+1} + 2 a_1 x^{r+1} = 0$$

Lowest power of x: x^r @ $n=0$. Indicial Eqn:

$$r(r-1) + 4r + 2 = 0.$$

$$\Rightarrow r^2 + 3r + 2 = 0 \Rightarrow \boxed{r = -1, -2}.$$

~~@ r=2~~ Comparing coefficients of both sides in (1):

x^{n+r}: $[(n+r)(n+r+3) + 2] a_n + a_{n-2} = 0.$

$$\Rightarrow a_n = \frac{-a_{n-2}}{(n+r)(n+r+3)+2}$$

$$\Rightarrow \boxed{a_{n+2} = \frac{-a_n}{(n+r+2)(n+r+5)+2}} \quad [\cancel{n=2,3,\ldots}]$$

x^{r+1}: $a_1 [(r+1) r + (1+r) + 2] = 0.$

$$\Rightarrow a_1 (r^2 + 2r + 3) = 0. \Rightarrow \boxed{a_1 = 0} \quad [r^2 + 2r + 3 > 0]$$

Largest indicial root: $r = -1$.

$$\therefore a_{n+2} = \frac{-a_n}{(n+1)(n+4)+2}$$

$n=0$: $a_2 = -\frac{a_0}{6}$.

$n=1$: $a_3 = \frac{-a_1}{12}$

$n=2$: $a_4 = \frac{-a_2}{20} = \frac{a_0}{6\cdot 20} = \frac{a_0}{120}$.

$n=3$: $a_5 = -\frac{a_3}{4\cdot 7+2} = \frac{-a_3}{30} = \frac{a_1}{12\times 30}$

$$\therefore y_1(x) = x^{-1}\sum_{n=0}^{\infty} a_n x^{n+}$$

$$= x^{-1}\left[a_0\left(1-\frac{x^2}{6}+\frac{x^4}{120}-\dots\right)+a_1\left(x-\frac{x^3}{12}+\frac{x^5}{12\times 30}-\dots\right)\right]$$

$$= x^{-1}\left[a_0\left(1-\frac{x^2}{3!}+\frac{x^4}{5!}-\dots\right)+a_1\left(x-\frac{2x^3}{4!}+\frac{2x^5}{6!}-\dots\right)\right]$$

$$= a_0\left(\frac{1}{x}-\frac{x}{3!}+\frac{x^3}{5!}-\dots\right)+a_1\left(1-\frac{2x^2}{4!}+\frac{2x^4}{6!}-\dots\right)$$

$a_1 = 0$

$$\therefore y_1(x) = a_0\left(x^{-1} - \frac{x}{3!}+\frac{x^3}{5!}-\dots\right)$$

$$= \frac{a_0}{x^2}\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\dots\right) = a_0\frac{\sin x}{x^2}$$

$$\therefore \boxed{y_1(x) \sim \frac{\sin x}{x^2}} \quad \text{(Ans)}$$

(b) $$y_2(x) = y_1\int \frac{\exp\left[-\int P\,dx'\right]}{y_1^2}\,dx \qquad \left[P(x)=\frac{4}{x}\right].$$

$$= \frac{\sin x}{x^2}\int \frac{1}{x^4}\cdot\frac{x^4}{\sin^2 x}\,dx.$$

$$= \frac{\sin x}{x^2}\int \operatorname{cosec}^2 x\,dx = \frac{\sin x}{x^2}\cdot -\cot x$$

$$= -\frac{\cos x}{x^2}$$

$$\therefore \boxed{y_2(x) \sim \frac{\cos x}{x^2}} \quad \text{(Ans)}$$

(c) $$y_2(x) = k y_1 \ln x + \sum_{n=0}^{\infty} d_n x^{n-2} \qquad [r_2=-2]$$

$$y_2'(x) = k\left[y_1'\ln x + \frac{y_1}{x}\right] + \sum_{n=0}^{\infty}(n-2)\,d_n x^{n-3}$$

$$y_2''(x) = k\left[y_1''\ln x + \frac{2y_1'}{x} - y_1/x^2\right] + \sum_{n=0}^{\infty}(n-2)(n-3)\,d_n x^{n-4}$$

~~Ya~~ Substituting into the ODE:

$$k\left[x^2 y_1'' + 4x y_1' + (x^2+2) y_1\right]\ln x \quad [y_1 \text{ is a soln of ODE}]$$

$$+ k(2xy_1' - y_1) + k\,4y_1 +$$

$$\sum_{n=0}^{\infty}(n-2)(n-3)d_n x^{n-2} + 4\sum_{n=0}^{\infty}(n-2)d_n x^{n-2} + \sum_{n=0}^{\infty} d_n x^n + 2\sum_{n=0}^{\infty} d_n x^{n-2} = 0.$$

$\therefore\ 2xy_1' + 3y_1.$

$$= 2x\sum_{n=0}^{\infty}(n-1)a_n x^{n-2} + 3\sum_{n=0}^{\infty} a_n x^{n-1} \quad [\text{from the definitn of } y(x)]$$

$$= 2\sum_{n=0}^{\infty}(n-1)a_n x^{n-1} + 3\sum_{n=0}^{\infty} a_n x^{n-1} = \sum_{n=0}^{\infty}(2n+1)a_n x^{n-1}$$

Coefficient of x^{-1}:

$$\cancel{-2a_0} + ka_0 + 2d_1 - 4d_1 + 2d_1 = 0. \quad [\text{no } x^{-1} \text{ term on RHS}].$$

$$\Rightarrow \quad ka_0 = 0 \quad \Rightarrow \quad \boxed{k=0} \quad [a_0 \neq 0].$$

(Proved)

(d) Recurrence Relatn:

$$a_{n+2} = \frac{-a_n}{(n+r+2)(n+r+5)+2}$$

@ $r_2 = -2$:

$$a_{n+2} = \frac{-a_n}{n(n+3)+2} \quad ; \quad \cancel{n=1,2,3}, \ldots$$

Hence, it never blows up.

$n=0$: $\quad a_2 = -\frac{a_0}{2} = -\frac{a_0}{2!}$

$n=1$: $\quad a_3 = \cancel{-a} - \frac{a_1}{6} = 0 \quad [a_1 = 0].$

$n=2$: $\quad a_4 = -\frac{a_2}{12} = \frac{a_0}{24} = \frac{a_0}{4!}$

$n=4$: $\quad a_6 = -\frac{a_4}{30} = -\frac{a_0}{6!}$

$$\therefore\ y_2(x) = x^{-2}\sum_{n=0}^{\infty} a_n x^n = x^{-2}\left[a_0\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots\right)\right]$$

$$= a_0 \cos x / x^2$$

$\therefore \boxed{y_2(x) = \frac{\cos x}{x^2}}$ (Ans)

6. $y'' + 2xy' + 2y = 0$

(a) No special pts for this ODE @ finite x.

@ $x \to \pm\infty$:

Let $t = \frac{1}{x}$. $\Rightarrow dt = -\frac{1}{x^2}dx = -t^2 dx$.

$\Rightarrow \frac{dy}{dt} = -1/t^2 \frac{dy}{dx}$

$\Rightarrow y' = -t^2 y'_t$ $\quad \cancel{\Rightarrow \frac{d^2y}{dt^2} = [2t\frac{dy}{dx} + t^2\frac{d^2y}{dx^2}\cdot\frac{dx}{dt}]}$

$\Rightarrow y'' = -[2t y'_t + t^2 y''_t]\cdot\frac{dt}{dx}$ $\quad \cancel{\to -2t y'}$

$= +2t^3 y'_t + t^4 y''_t$

Substituting into the ODE:

$2t^3 y'_t + t^4 y''_t + \frac{2}{t}\cdot(-t^2 y'_t) + 2y_t = 0$

$\Rightarrow y''_t + 2\left(\frac{1}{t} - \frac{1}{t^3}\right)y'_t + \frac{2}{t^4}y_t = 0$.

$\therefore P(t) = \frac{1}{t} - \frac{1}{t^3} = \frac{t^2-1}{t^3} \to$ Pole of order 3 @ $t=0$.

$Q(t) = \frac{2}{t^4} \to$ Pole of order 4 @ $t=0$.

$\therefore t=0 \Rightarrow x \to \infty \to$ Irregular singular pt.

(b) $y(x) = \sum_{n=0}^{\infty} a_n x^n$

$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$

$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

Substituting into the ODE:

$\underbrace{\sum_{n=2}^{\infty} a_n n(n-1) x^{n-2}}_{n \to n+2} + 2\underbrace{\sum_{n=1}^{\infty} n a_n x^n}_{(n=0 \text{ gives } 0 \text{ anyways})} + 2\sum_{n=0}^{\infty} a_n x^n = 0$

$\Rightarrow \sum_{n=0}^{\infty} [(n+1)(n+2)a_{n+2} + 2a_n(n+1)]x^n = 0$

Comparing both sides:

$$(n+1)(n+2)a_{n+2} + 2a_n(n+1) = 0.$$

$$\Rightarrow (n+1)[(n+2)a_{n+2} + 2a_n] = 0.$$

$$\Rightarrow \boxed{a_{n+2} = -\frac{2a_n}{n+2}}$$

$n=0$: $a_2 = -a_0$.

$n=1$: $a_3 = -\frac{2a_1}{3}$.

$n=2$: $a_4 = -\frac{a_2}{2} = \frac{a_0}{2}$.

$n=3$: $a_5 = -\frac{2a_3}{5} = \frac{4}{15}a_1$

$$\therefore y(x) = \sum_{n=0}^{\infty} a_n x^n.$$

$$= a_0\left(1 - x^2 + \frac{x^4}{2} + \dots\right) + a_1\left(x - \frac{2x^3}{3} + \frac{4}{15}x^5\right) + \dots$$

Let a Even Sol^n: $a_1 = 0$; $a_0 = 1$.

$$y_1(x) = 1 - x^2 + \frac{x^4}{2} + \dots = \boxed{e^{-x^2} = y_1(x)} \quad \text{(Ans)}$$

Odd Sol^n: $a_0 = 0$; $a_1 = 1$.

$$\boxed{y_2(x) = \left(x - \frac{2x^3}{3} + \frac{4}{15}x^5 + \dots\right)} \quad \text{(Ans)}$$

The Sol^n is valid in entire $\mathbb{R}$. $(x \in \mathbb{R})$.

Convergence:

$$\frac{a_{n+2}}{a_n} = \frac{-2}{n+2}.$$

For $n \to \infty$: $\frac{a_{n+2}}{a_n} = \frac{-2}{n} < 0$.

Thus, the series converges. (Ans)

(7) Bessel's Fn of order 0.